TEJA MADIRE

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PROFESSIONAL SUMMARY

Data Engineer with 3 years of experience building production-grade data pipelines and lakehouses across real estate, water management, and AI/ML platform clients — Medallion Architecture, Apache Iceberg, PySpark, AWS EMR, AWS Glue, and Apache Airflow on AWS.

TECHNICAL SKILLS

Programming Languages: Python (Pandas, PySpark), SQL (Advanced)
Workflow Orchestration: Apache Airflow 3.0 (MWAA, DAG authoring, EmrAddStepsOperator, SLA monitoring)
Big Data & Cloud (AWS): PySpark, Apache Spark, Apache Kafka, Apache Iceberg, AWS EMR, AWS S3, AWS Glue, Amazon Kinesis Firehose, AWS Lambda, Amazon Athena, Amazon QuickSight
Databases & Warehousing: PostgreSQL, MySQL, Amazon Redshift
Data Modelling & Transformation: Medallion Architecture, Star Schema, dbt Cloud
DevOps & CI/CD: Git, GitHub, GitHub Actions, Docker
Certifications: AWS Solutions Architect Associate, AWS Data Analytics Specialty

PROFESSIONAL EXPERIENCE

Associate Data Engineer | Mactores *Feb 2024 – Present*

Project 1: Data Lakehouse Foundation – Real Estate Client (AWS EMR, PySpark, Apache Iceberg, Airflow MWAA)

- Designed and built a config-driven PySpark ETL engine powering 51+ entity pipelines from a single codebase on AWS EMR; each pipeline independently configurable for schema, partitioning, and DQ rules — reducing new pipeline onboarding time by ~70%.
- Built 9 Gold layer dimension and fact table jobs using PySpark on EMR; the most complex job (dim_member) joins 10+ Silver tables into a single 100+ column analytical table, with auto-detection logic to switch between full-load and incremental-load patterns based on target table state.
- Authored 52+ Airflow (MWAA) DAGs to orchestrate Silver and Gold EMR Spark jobs on scheduled intervals; implemented retry logic, SLA alerts, and failed_states validation on EmrStepSensor — eliminating silent pipeline failures.
- Implemented a PyDeequ-based data quality framework splitting records into pass/fail streams per pipeline run; wrote failure metrics to S3 for monitoring — reduced data quality incidents by flagging issues before they reached the Gold layer.

Project 2: Email Data Processing Optimization – Next Best Action Platform (AWS Glue, PySpark, Delta Lake)

- Investigated a 3+ hour AWS Glue PySpark job processing 2.69 billion email log records from Delta Lake on S3; identified the root cause as raw HTML/Liquid template content being loaded as-is into the target table, discussed requirements with the client, and implemented a Spark-native HTML/template cleanup transformation with targeted filters — combined with a broadcast join for the 6.28M-row content lookup and Spark shuffle/memory tuning — reducing runtime from 3+ hours to under 10 minutes and producing 400GB of compressed Snappy Parquet on S3.

Project 3: Enterprise Data Lake – Water Management Solutions Client (AWS EMR, AWS Glue, S3, Redshift, Amazon QuickSight)

- Designed and deployed PySpark batch pipelines on AWS EMR to ingest, clean, and transform high-volume IoT sensor data (water management); implemented incremental loads and schema validation — ensuring consistent data availability for downstream reporting.
- Wrote efficient SQL queries within PySpark to extract and aggregate water usage KPIs, filtration efficiency, and irrigation performance metrics; loaded curated outputs into Amazon Redshift to support scalable data warehousing and downstream analytics.
- Monitored pipeline performance end-to-end via CloudWatch; proactively resolved data quality issues and optimised Spark job execution — reducing average job runtime and preventing SLA breaches on daily reporting cycles.
- Collaborated with cross-functional stakeholders to understand reporting requirements; delivered Amazon QuickSight dashboards and maintained technical documentation covering data flows, transformation logic, and pipeline design.

Data Engineer Intern | Mactores *May 2023 – Feb 2024*

- Attained AWS Solutions Architect Associate and AWS Data Analytics Specialty certifications; applied cloud architecture and ETL/ELT best practices directly to team pipeline work.
- Assisted senior data engineers in building and optimizing ETL workflows on AWS; gained hands-on exposure to Python and SQL pipeline development, debugging and troubleshooting data issues, and version control with Git and GitHub.
- Independently built Python and SQL data engineering projects using Pandas, and AWS services; practised data cleaning, transformation, warehousing concepts, and basic CI/CD practices using GitHub Actions.
- Deployed the CUDOS Dashboard (AWS Cloud Intelligence Dashboards) for a client to analyze cost and usage across their AWS organization; utilized AWS Glue, Lambda, Athena, and CloudFormation to extract and store 65GB of cost and usage reports with tag-based breakdowns (project, environment, department).
- Developed budgeted vs. actual spend analysis by integrating AWS Budget API data into QuickSight; implemented a QuickSight Q topic enabling business users to query cost and usage data using natural language.

PORTFOLIO & PERSONAL PROJECTS

RetailPulse — Unified Retail Analytics Lakehouse | Python, PySpark, AWS EMR, Apache Iceberg, Airflow 3.0 MWAA, dbt Cloud, Kinesis Firehose, Athena, QuickSight
github.com/teja-madire-BigData/RetailPulse-Lakehouse

Built end-to-end batch lakehouse on AWS — 6 Lambda extractors (Faker synthetic + ExchangeRate-API v6 + Open-Meteo) route records via Kinesis Firehose into append-only Apache Iceberg Bronze tables; two-seed Faker strategy (seed=42 catalogue + datetime event seed) ensures stable FK joins while generating genuine incremental deltas each run.

Implemented 6 PySpark Silver jobs on EMR 7.12: SCD Type 1 MERGE INTO upserts for products, users, and sellers using Window deduplication; idempotent append + DQ (assert checks on row counts, nulls, valid statuses) for orders, FX rates, and weather; orchestrated by 6 independent Airflow 3.0 MWAA DAGs using EmrAddStepsOperator with logical_date − timedelta(days=1) partition targeting.

Designed Gold star schema (7 dbt Cloud models: 5 dims + 2 facts) using dbt-athena adapter with Apache Iceberg materialisation; key metric order_total_usd normalises 6 currencies (USD/EUR/GBP/INR/AED/SGD) to USD via daily FX rates — enabling accurate cross-currency revenue reporting across all channels.

YouTube Trending Data Pipeline | Python, YouTube Data API v3, AWS S3, AWS Glue, Amazon Athena, Apache Airflow, Amazon QuickSight
github.com/teja-madire-BigData/YouTube_Data_Analysis_Pipeline_From_Extraction_to_Visualization

Built a Python-based daily ingestion pipeline using YouTube Data API v3 — extracted trending video data (title, category, view/like/comment counts) for 5 countries (Brazil, India, Indonesia, Mexico, US); flattened nested JSON, mapped category IDs, and landed per-country CSV files to S3.

Orchestrated via Airflow DAG with 5 parallel PythonOperator tasks (one per country) converging into a final Athena query task; AWS Glue crawled S3 for automatic schema inference enabling ad-hoc SQL analysis of trending categories and engagement metrics via Athena and QuickSight.

CERTIFICATIONS

AWS Solutions Architect Associate Amazon Web Services	**AWS Data Analytics Specialty** Amazon Web Services

EDUCATION

Master of Computer Applications (MCA) — Kakatiya University, Warangal, India *Dec 2020 – Sep 2022*
Bachelor of Computer Applications (BCA) — Siddhartha Degree and PG College, India *Jul 2017 – Sep 2020*